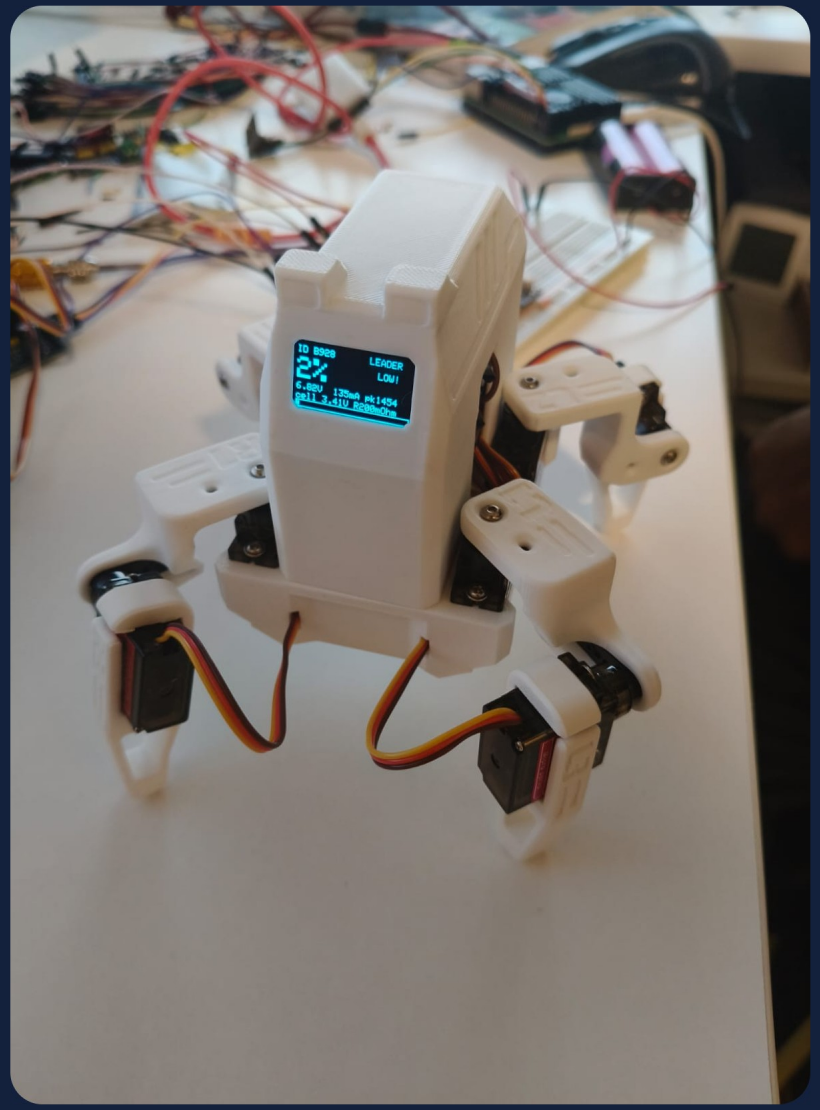
WIRELESS ROBOTICS PROJECT

Sesame Robots

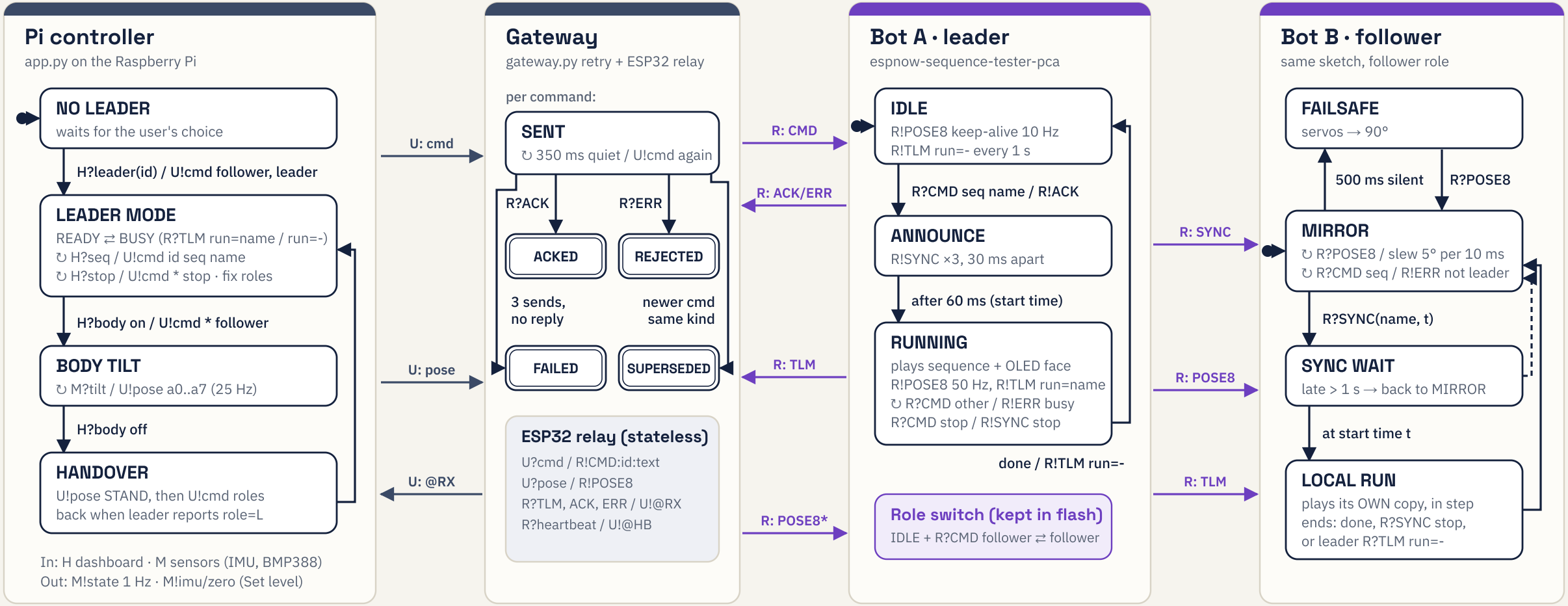
Two four-legged robots that move in step, controlled from a Raspberry Pi ground station.



Abubakr and Mustafa

Communicating state machines

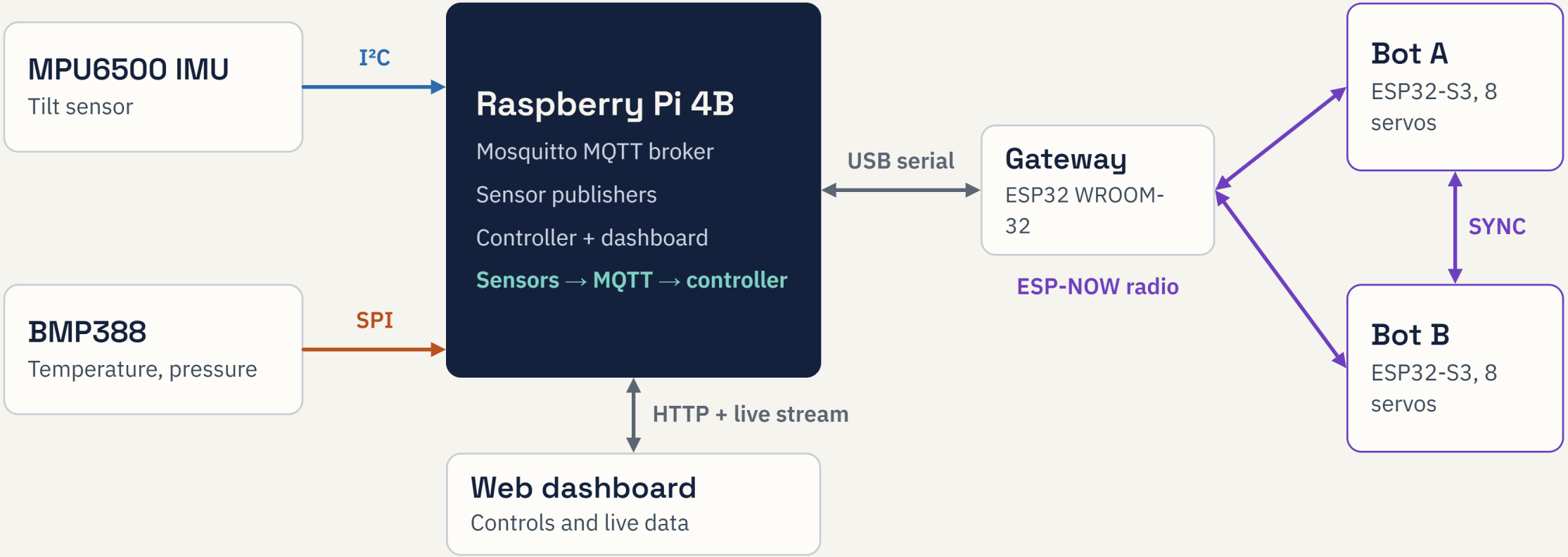
Each machine runs on its own and reacts only to messages, so a lost packet never stalls a bot.



c!m sends m on channel c, **c?m** receives it, ↻ stays in the state. **H** HTTP dashboard **M** MQTT **U** USB serial **R** ESP-NOW broadcast, ch 1 *Pi streams POSE8 only in body tilt.

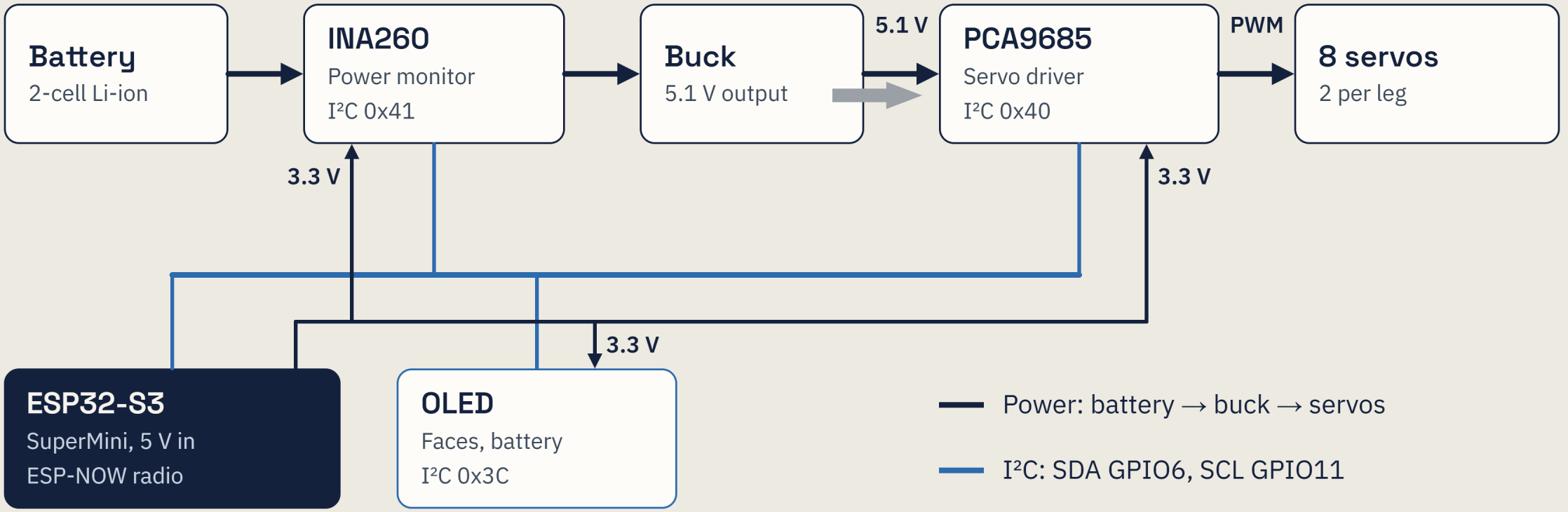
System overview

The Pi reads the sensors and runs the dashboard; one USB radio links it to both bots.



Bot schematic

One battery powers the whole bot; one I²C bus links its three chips to the ESP32-S3.



All grounds are common. The buck also feeds the ESP32's 5 V pin; the chips' logic runs from its 3.3 V pin.

Pinouts

ESP32-S3 SuperMini (each bot)

Pin	Connects to
GPIO6	I²C SDA: PCA9685, INA260, OLED
GPIO11	I²C SCL: same three chips
5V	Buck output (5.1 V)
3V3	Logic power for the three chips
GND	Common ground
USB-C	Flashing and serial monitor

I²C addresses: PCA9685 0x40 · INA260 0x41 · OLED 0x3C

Raspberry Pi 4B (ground station)

Pin	GPIO	Connects to
1	3V3	IMU + BMP388 power
3	GPIO2 SDA	MPU6500 SDA (I²C)
5	GPIO3 SCL	MPU6500 SCL (I²C)
6	GND	Sensor ground
19	GPIO10 MOSI	BMP388 SDI (SPI)
21	GPIO9 MISO	BMP388 SDO (SPI)
23	GPIO11 SCLK	BMP388 SCK (SPI)
29	GPIO5	BMP388 CS (SPI)
USB	–	Gateway ESP32 (serial)

The gateway (ESP32 WROOM-32) needs no wiring beyond its USB cable to the Pi.

Communication links

ON THE BOARDS

I²C

Each bot: the ESP32-S3 drives the servo driver, power monitor and OLED at 400 kHz. On the Pi: the IMU.

ON THE PI

SPI

The Pi reads the BMP388 temperature and pressure sensor once a second.

OVER THE AIR

ESP-NOW

Wi-Fi radio with no router. Commands and telemetry between gateway and bots; the leader starts the follower in step.

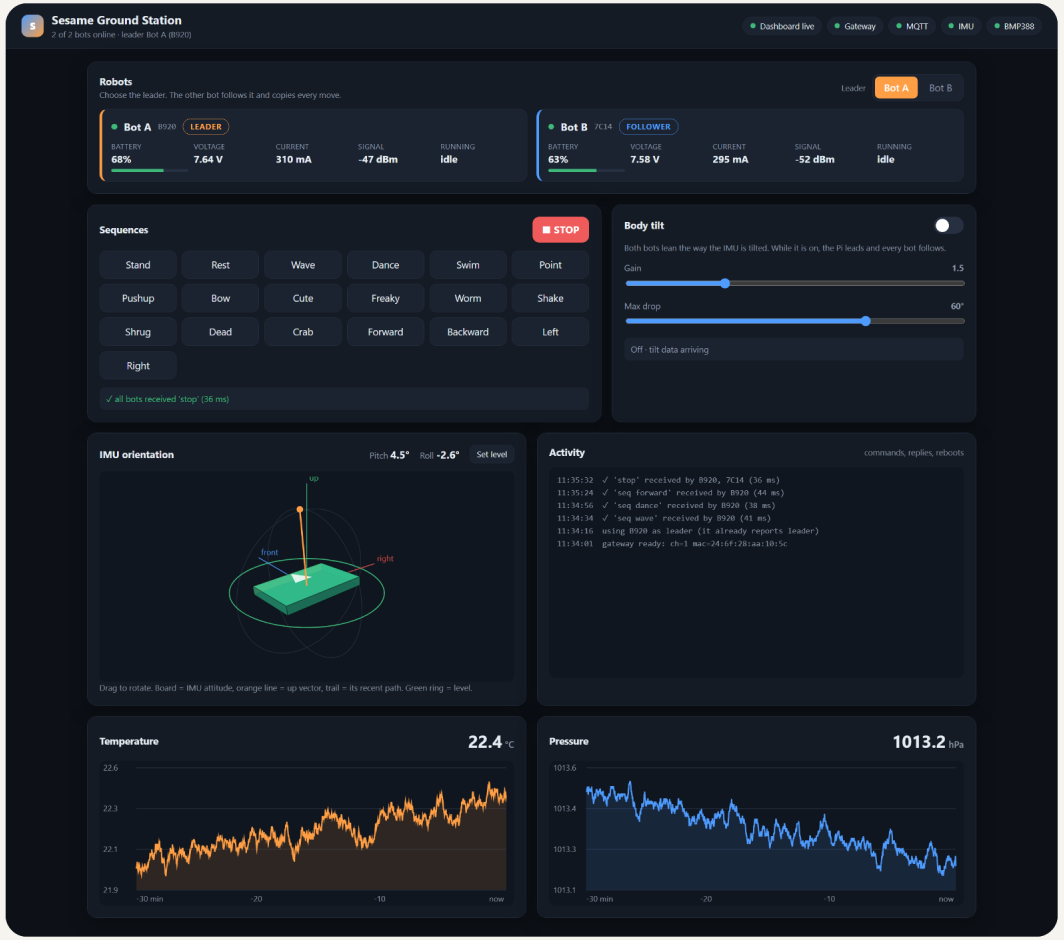
INSIDE THE PI

MQTT

A message broker. Sensor scripts publish tilt, temperature and pressure; the controller subscribes.

Also: **USB serial** from the Pi to the gateway, and **HTTP with a live stream** to the dashboard.

Ground station dashboard



Pick the leader

One click; the other bot follows. The choice survives reboots.

Run sequences

19 routines plus STOP, with a delivery check for every command.

Body tilt

Both bots lean with the IMU in real time.

Live data

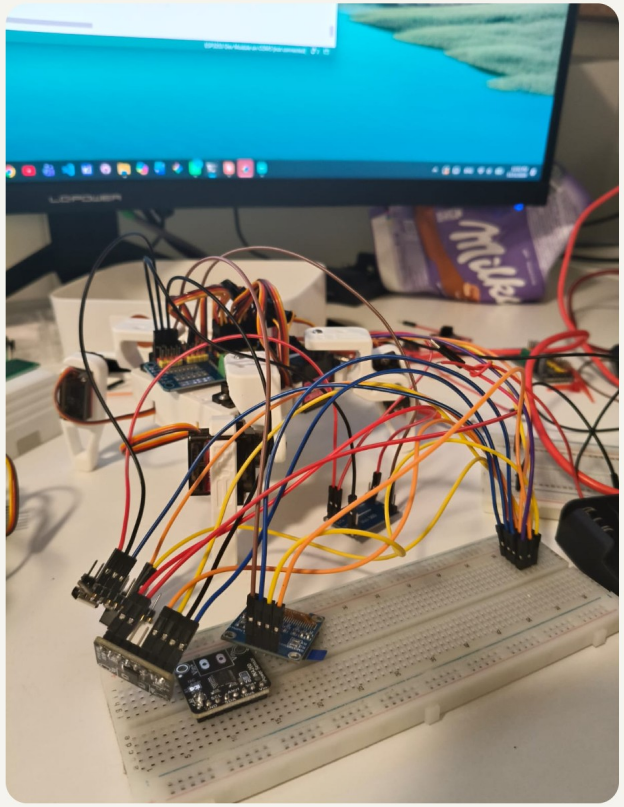
Battery, signal, IMU attitude, temperature and pressure.

Screenshot shown with sample data.

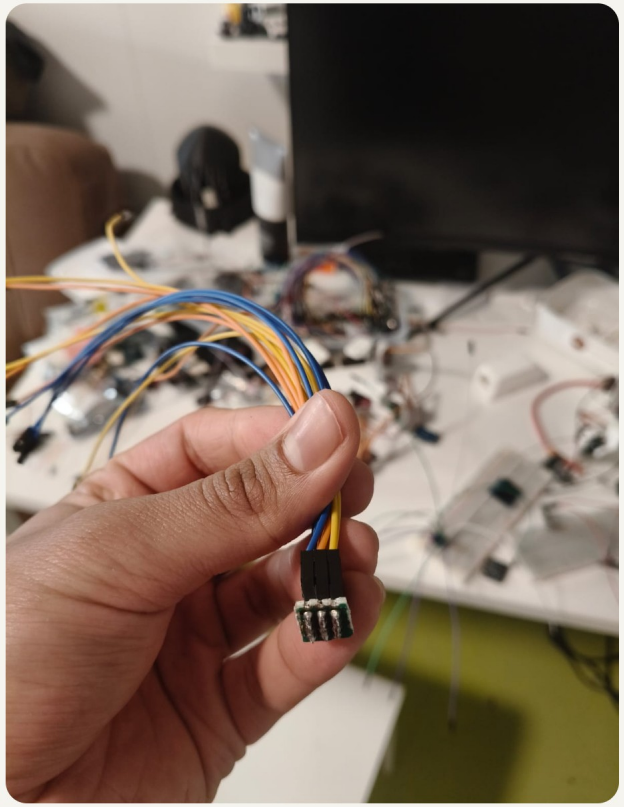
Wiring the I²C bus

Four boards share two signal wires: SDA and SCL.

The breadboard prototype proved the bus worked. A hand-soldered bridge then joined the bus wires so every chip connects at one point inside the body.



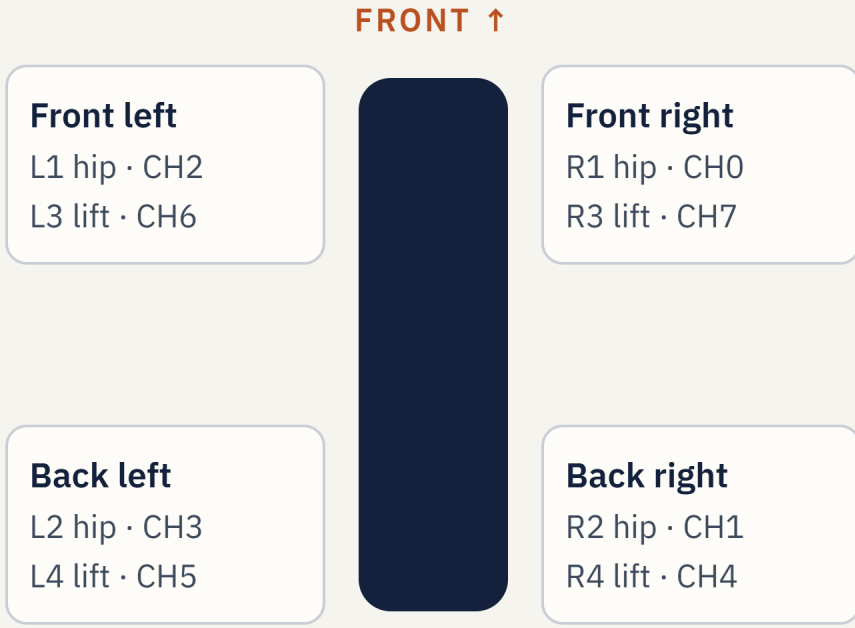
Breadboard prototype of the bus



Soldered I²C bridge

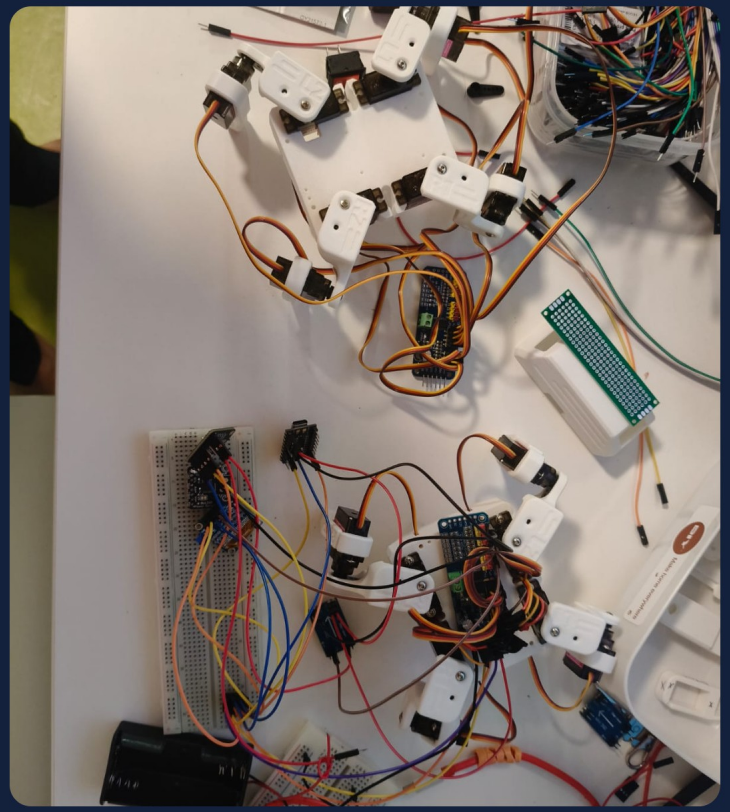
PCA9685 channels to joints

Channel	Joint	Leg	Type
CH0	R1	Front right	Hip
CH1	R2	Back right	Hip
CH2	L1	Front left	Hip
CH3	L2	Back left	Hip
CH4	R4	Back right	Lift
CH5	L4	Back left	Lift
CH6	L3	Front left	Lift
CH7	R3	Front right	Lift



50 Hz servo pulses of 500–2500 µs. Hips swing a leg forward and back; lifts raise and lower it. Channels 8–15 are unused.

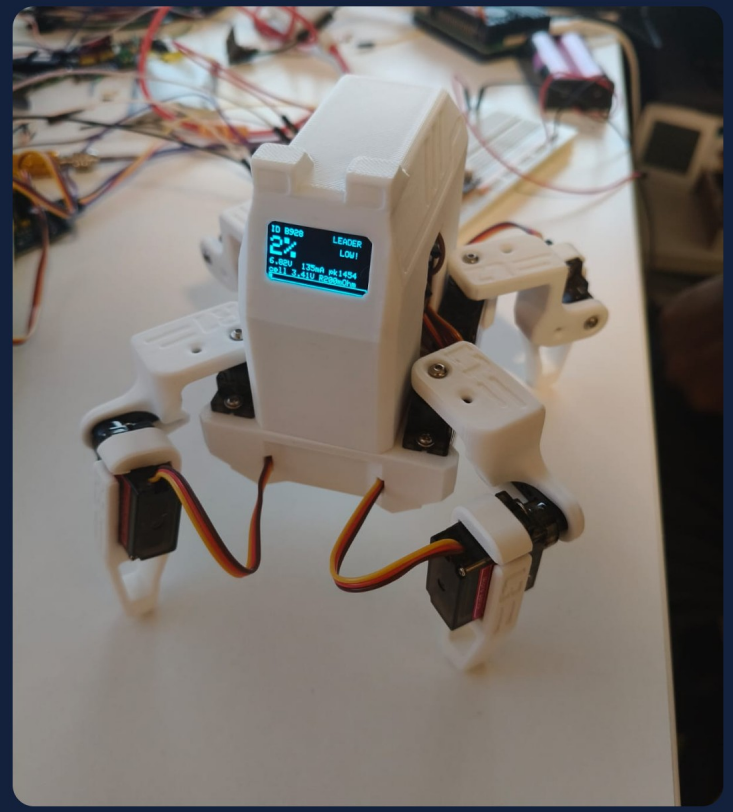
From prototype to finished bot



Both bots during assembly



First body, before the head extension



Finished bot, OLED on the battery screen